

Radii systematic for various DFT in Ca, Ni, Sn, and Pb chains

Internal notes: FAU – MSU

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The archive delivered together with these internal notes contains a collection of charge r.m.s. radii in the chain of Ca-, Ni-, Sn-, and Pb-isotopes from proton dripline to neutron dripline (and a bit beyond) computed with a variety of DFT parametrizations. The results come in the following files, one separate file for each functional:

`collect.SV-min:`

fit Skyrme functional to semi-magic nuclei [1]

`collect.SV-min-HFBcut5`

SV-min where pairing is refitted with HFB

`collect.UNEDF1-BCScut5-rho16:`

fit Skyrme functional to deformed nuclei, BCS pairing refitted [2]

`collect.UNEDF1.HFB-rho16:`

UNEDF1 with HFB pairing refitted

`collect.Fystd:`

Fy(std) = fit Fayans functional to data set of SV-min [3]

`collect.FyDr:`

Fy(Δr) = extend Fy(std) to include radius differences in Ca [3]

`collect.FyDrHFB:`

Fy(Δr ,HFB) = extend Fy(Δr) by data from ^{36}Ca and use HFB pairing [4]

`collect.FyIVP-stab05:`

Fy(IVP) = extend Fy(Δr ,HFB) by different proton-neutron pairing [5]

`exp-radii-CaNiSnPb.dat:`

collection of experimental r.m.s. charge radii

Supposing that all considered nuclei are spherical, the calculations were done with a spherical 1D code. The now following figures should give an impression what the files contain and to address possible limitations of a spherical mean-field approach.

Figure 1 summarizes all results for the r.m.s. radii. They separate into two almost distinctive groups of predictions. At the one side, the four Skyrme functionals plus the Fayans functional Fy(std), and at the other side, the three remaining Fayans functionals

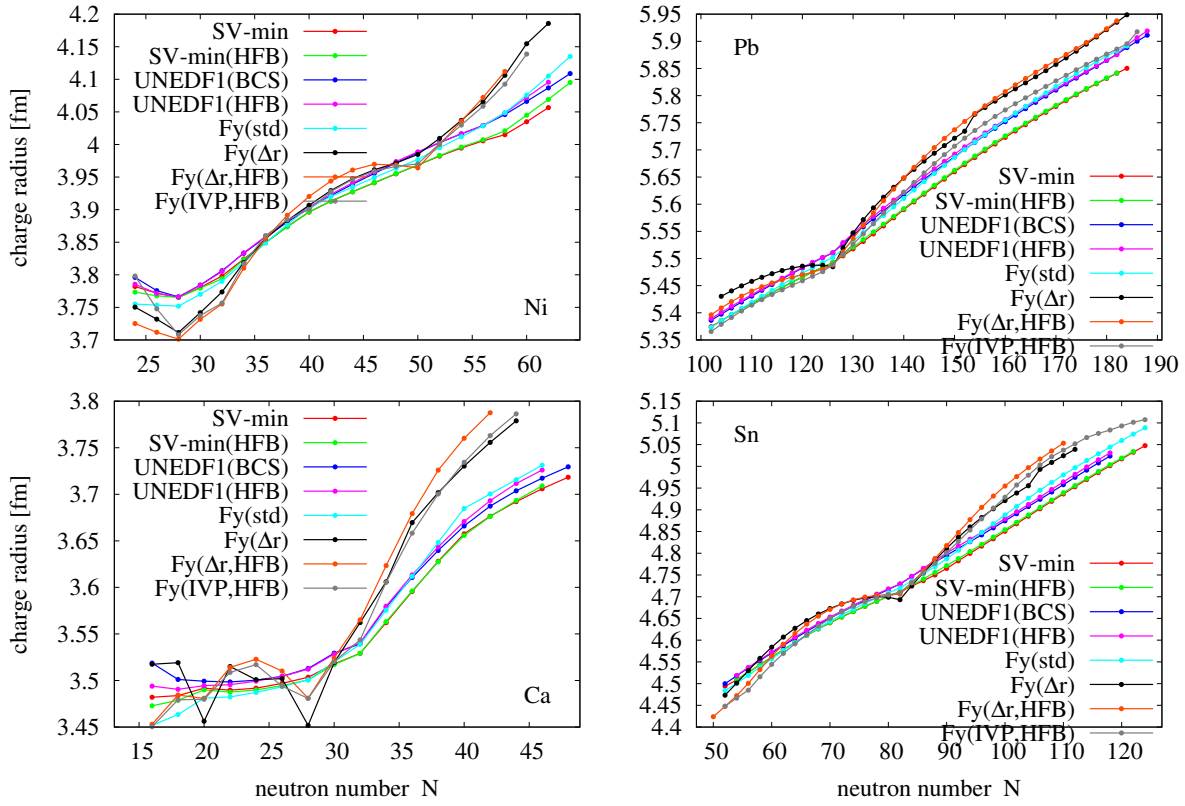


Figure 1: Charge r.m.s. radii for the four isotopic chains and all DFT functionals under consideration. Only result for isotopes within the driplines of each DFT have been plotted.

$Fy(\Delta r)$, $Fy(\Delta r, HFB)$, and $Fy(IVP)$. The latter show a strong enhancement for mid-shell Ca isotopes in agreement with data (not shown) which is achieved by the gradient pairing term in the Fayans functional and by including difference radii in Ca as fit data.

Ca isotopes, pairing properties

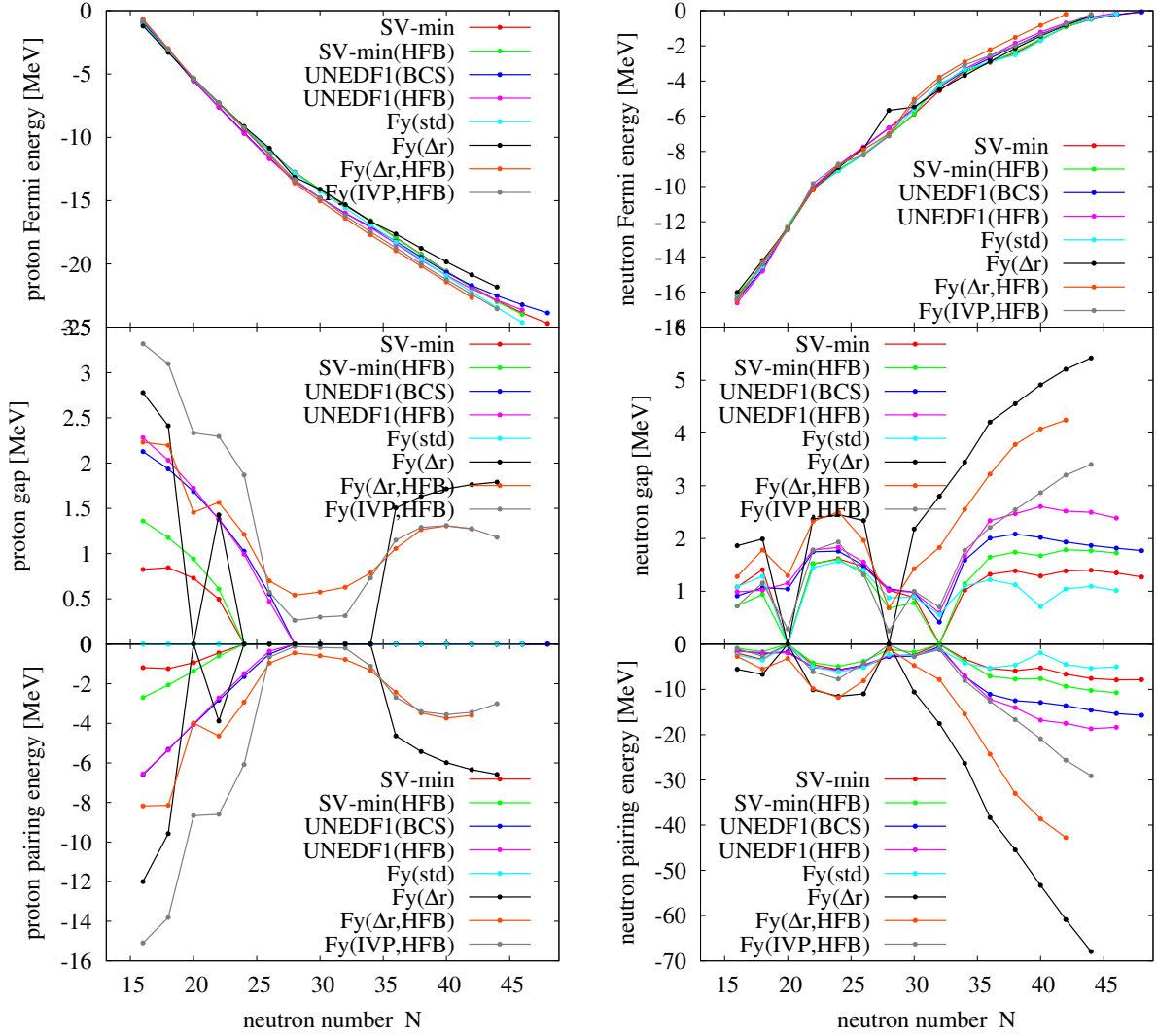


Figure 2: Pairing properties (pairing energy, average gap, Fermi energy) for protons (left) and neutrons (right) along the chain of Ca isotopes for all DFT functionals under consideration. Only result for isotopes within the driplines of each DFT have been plotted.

Figures 2–5 collect all information on pairing properties. The Fermi energies (upper panels) develop with expected trends. Protons are less bound with shrinking neutron number and neutrons become less bound with increasing neutron number. Both trends are terminated with the respective dripline. Pairing energies (lower panels) and $u_\alpha v_\alpha$ -averaged pairing gaps (middle panels) contain similar information with different bias. The gaps indicate how far the occupation distribution reaches above the Fermi energy. This together with the position of the Fermi energy allows to estimate the impact of the continuum. It grows obviously toward both driplines. This puts a warning sign that mere BCS calculations become unreliable for the very exotic nuclei near driplines. As a

rule-of-thumb, one should be careful with all isotopes having Fermi energy less than 2 MeV below continuum threshold.

It is also interesting to note that proton pairing does not always vanish although we are dealing with magic proton numbers throughout. Proton pairing sets is regularly coming up and increasing the farther away from the regime of stable nuclei. A minor difference is seen for Fy(Δr ,HFB) and Fy(IVP) which both use stabilized pairing keeping always some minimal amount of pairing. Thus we see some small pairing also for stable nuclei. But even here the growth toward driplines is obvious.

The discussion is continued with $\hookrightarrow$ figure 6 below.

Ni isotopes, pairing properties

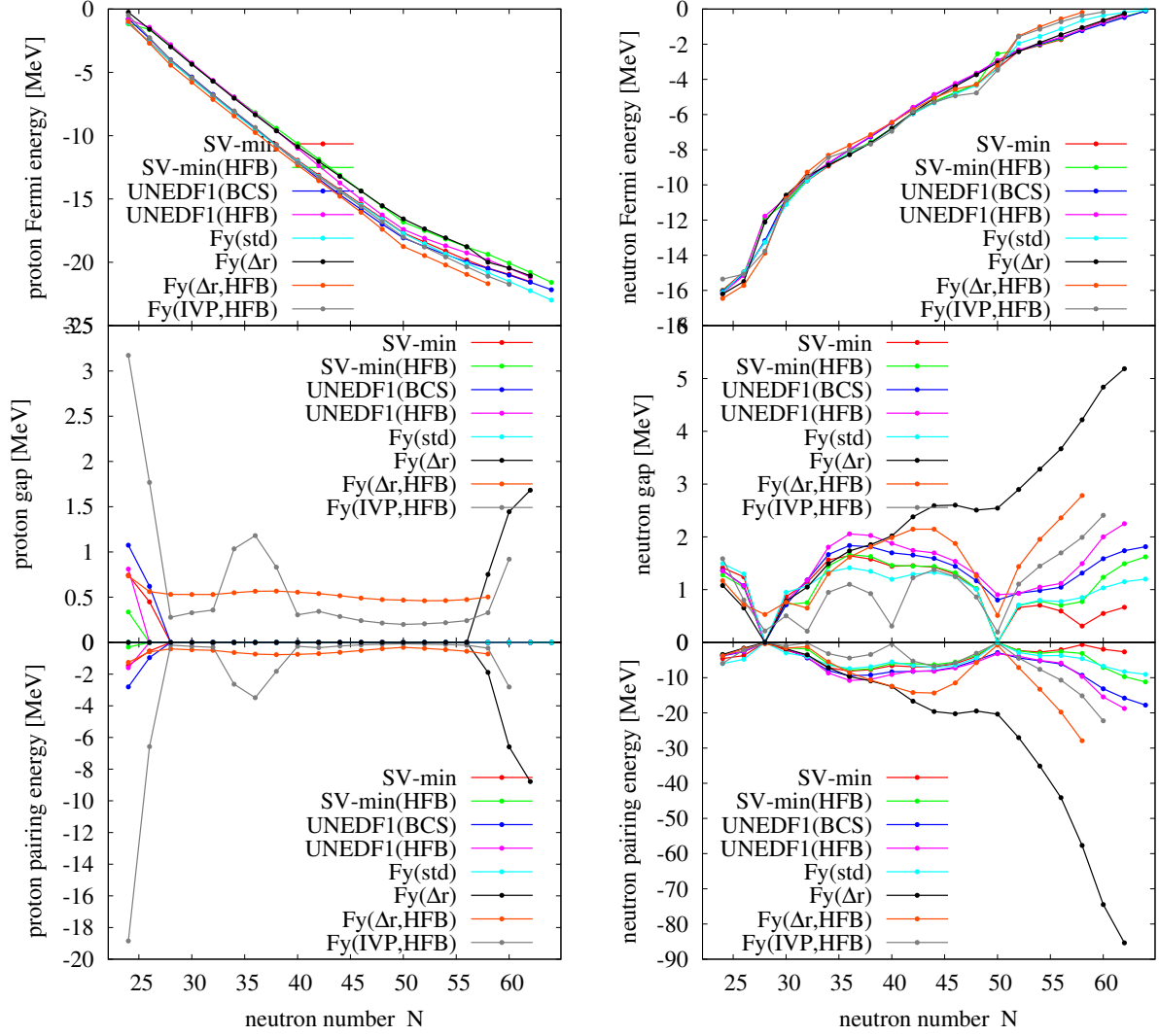


Figure 3: Pairing properties (pairing energy, average gap, Fermi energy) for protons (left) and neutrons (right) along the chain of Ni isotopes for all DFT functionals under consideration. Only result for isotopes within the driplines of each DFT have been plotted.

Sn isotopes, pairing properties

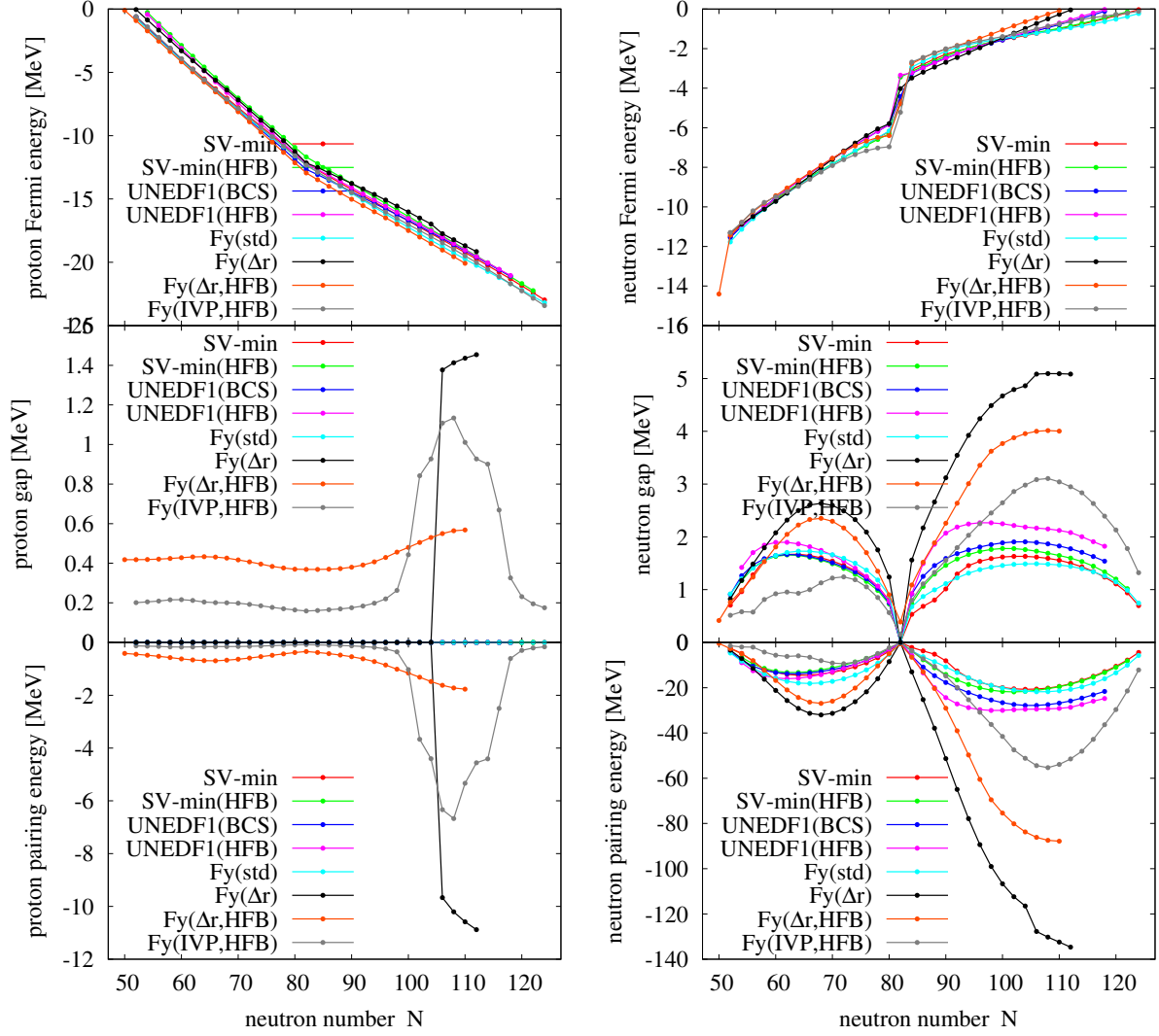


Figure 4: Pairing properties (pairing energy, average gap, Fermi energy) for protons (left) and neutrons (right) along the chain of Sn isotopes for all DFT functionals under consideration. Only result for isotopes within the driplines of each DFT have been plotted.

Pb isotopes, pairing properties

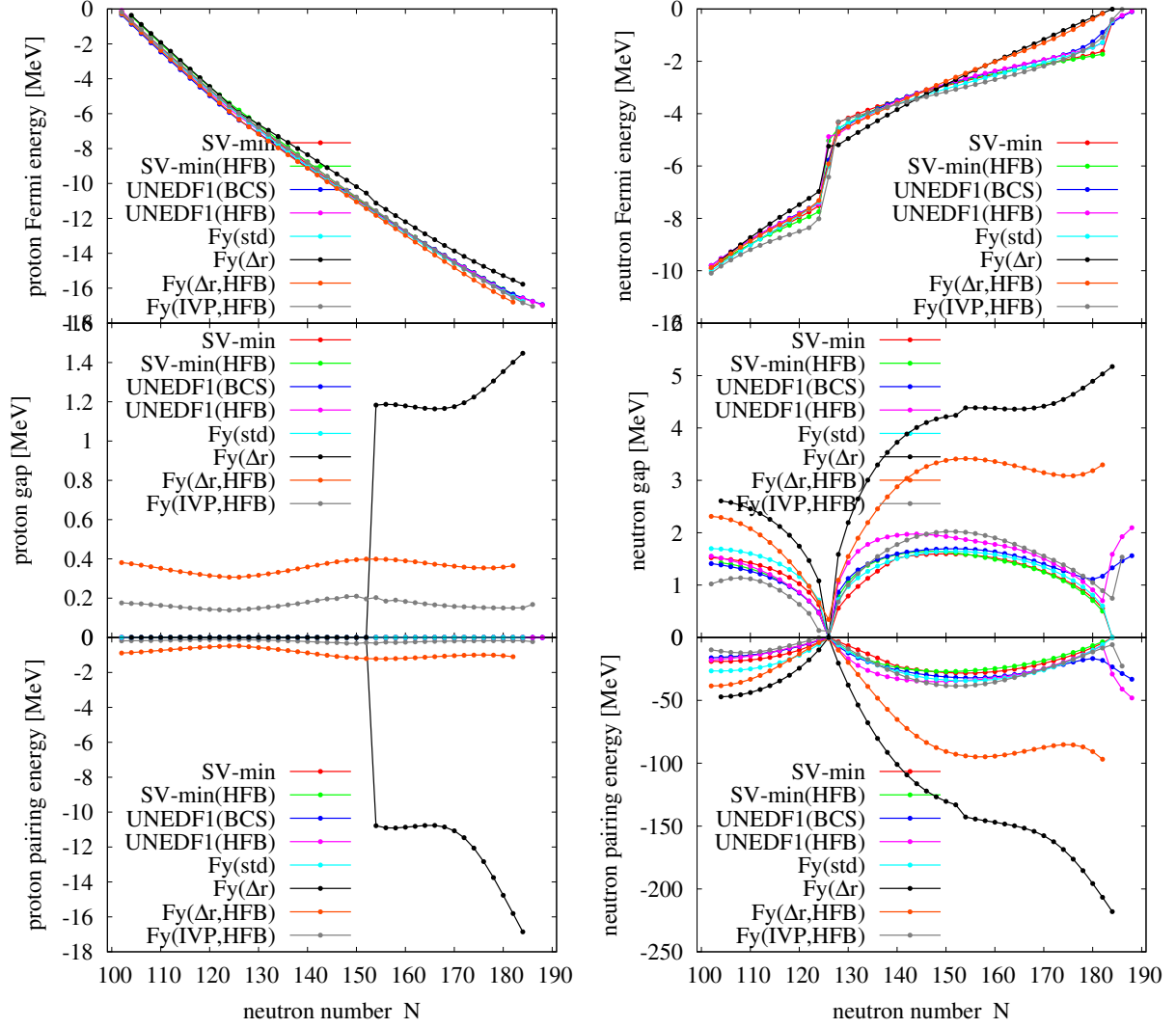


Figure 5: Pairing properties (pairing energy, average gap, Fermi energy) for protons (left) and neutrons (right) along the chain of Pb isotopes for all DFT functionals under consideration. Only result for isotopes within the driplines of each DFT have been plotted.

curvature of collective PES as measure for softness, SV-min

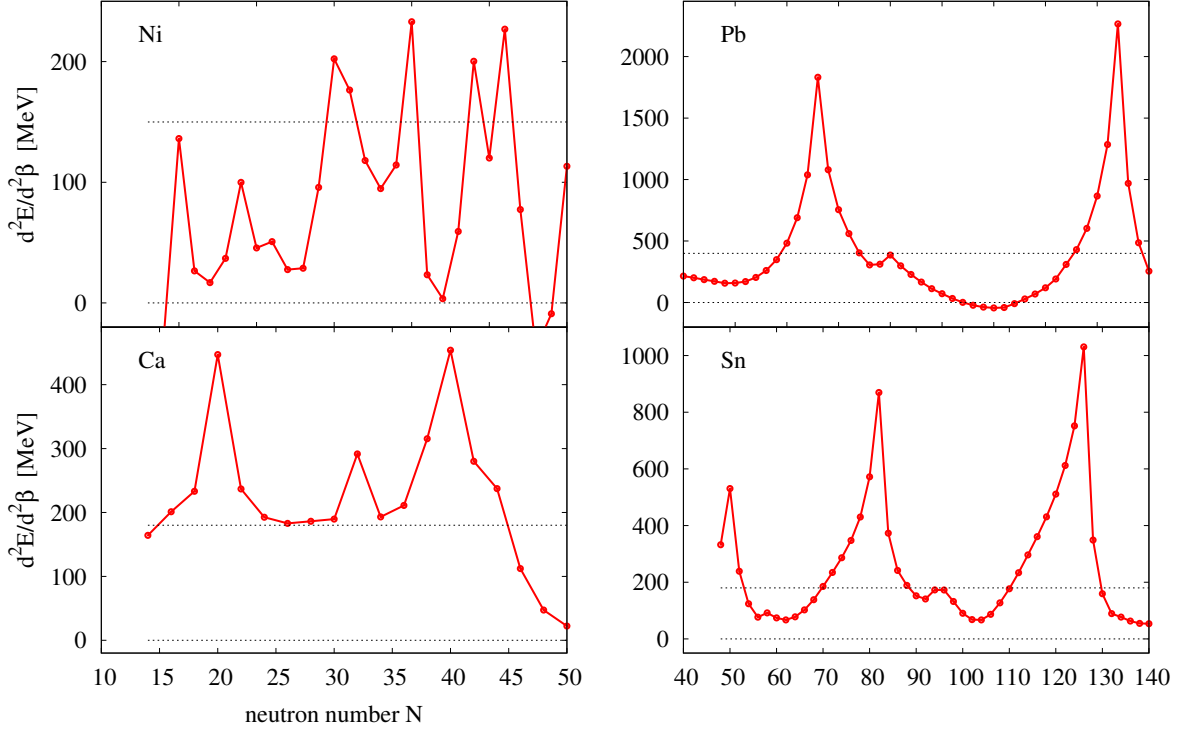


Figure 6: The softness of nuclei estimated from the curvature $\partial_\beta^2 E(\beta)$ of the collective deformation energy $E(\beta)$ at the spherical point. Results computed with SV-min are shown for the four isotopes under consideration. Each panel contains two faint horizontal line. One line indicates curvature zero to distinguish stable spherical minima (positive curvature) from deformed nuclei. The other horizontal line indicates a “threshold” below which we have to expect considerable correlation corrections to the radii. The height of this line is chosen by experience.

Fermi energies close to continuum threshold are one limitation which, however, applies only when using BCS. There is a more principle limitation of mean-field calculations coming from collective vibrational correlations in soft nuclei [6]. We need to check that for the present selection of semi-magic isotopic chains. As a rough measure of softness, we compute the curvature $\partial_\beta^2 E(\beta)$ of the quadrupole deformation energy surface. The results are shown in figure 6. They are shown only for SV-min. This should serve to demonstrate general trends. The pattern can change quantitatively for other functionals.

The maxima at doubly-magic nuclei are obvious. These represent surely rigid vibrators together with their vicinity. The mean-field approximation is valid here. Farther away from doubly-magic configurations the curvature can become suspiciously small. The upper horizontal faint lines indicate a limit below which experience shows that correlation corrections to radii become considerable. This is also a regime which has to be taken with care.

The lower horizontal line at curvature zero indicates the point below which spherical

nuclei are unstable against deformation. It is interesting to see that this happens rarely, only for very neutron rich Pb isotopes.

References

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